GMI German-Malaysian Institute

DIPLOMA PROGRAMME

LAB WORK 1

Academic Period: January 2026

Program	:	**CYBER SECURITY TECHNOLOGY (CBS)**
Course	:	**PENETRATION TESTING**
Code	:	**CBS 2353**
Total Marks	:	**20 MARKS**

Title	:	**INFORMATION GATHERING** [Shodan Searches]

Student's Name	:	Abdul Hariz Bin Mohd Sarnudin
TTO's Name	:	Ts. Noormelah Binti Shamsul Anuar
Semester/Trade/Group	:	SEM 4 / DCBS / CBS 1 (CBS24070208)
Date	:	10/2/2026

Assessment Specification Table (CBS)

Lab work

Raw Marks:	100
Percentage (in Doc 01):	30

No	Assessment Criteria	Full Marks	Score (if applicable)	Student's Marks	CLO	Level of Difficulty	DK, DP, NA (ETAC Programme Only)	Comments	PLO GMI: PLO CBS 9 PLO - Start Intake July 2023	PLO GMI: PLO MBOT 9 PLO
1	Understanding of Lab Objectives	4			2	3			2	2
2	Demonstration of Labwork Execution	4			2	3			2	2
3	Accuracy of Findings / Results	4			2	3			2	2
4	Fluency and Clarity of Explanation	4			2	3			2	2
5	Question & Answer (Q&A)	4			2	3			2	2
									#N/A	#N/A
									#N/A	#N/A
									#N/A	#N/A
									#N/A	#N/A
	Total Full Marks	**20**	**Total Student's Marks**							

SUMMARY

Note:
FM - Full Marks,
StM - Student's Mark

	CLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM	PLO GMI: PLO CBS 9 PLO - Start Intake July 2023 PLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM	PLO GMI: PLO MBOT 9 PLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM
Total	CLO 1	0		0.0		PLO 1	0		0.0		PLO 1	0		0.0	
	CLO 2	20		6.0		PLO 2	20		6.0		PLO 2	20		6.0	
	CLO 3	0		0.0		PLO 8	0		0.0		PLO 8	0		0.0	
	CLO 4	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	CLO 5	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	CLO 6	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	Total Marks	**20**		**6.0**											

Signature

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NO.	CRITERIA	EXCELLENT [4]	GOOD [3]	SATISFACTORY [2]	WEAK/ NOT MET [1]	MARKS
INDIVIDUAL COMPONENT (TOTAL MARK = 20)						
1	**Understanding of Lab Objectives**	Clearly explains the purpose, scope, and objectives of the lab work	Explains objectives with minor gaps	Basic understanding of objectives	Objectives misunderstood or not explained	
2	**Demonstration of Labwork Execution**	Demonstrates lab tasks correctly, confidently, and independently	Demonstration mostly correct with minor errors	Demonstration partially correct or requires guidance	Unable to demonstrate lab tasks correctly	
3	**Accuracy of Findings / Results**	Findings are accurate, relevant, and clearly linked to the task	Mostly accurate findings with minor errors	Limited or partially correct findings	Unable to demonstrate lab tasks correctly	
4	**Fluency and Clarity of Explanation**	Explanation is clear, fluent, well-structured, and uses correct technical terms	Generally clear with minor language or structure issues	Explanation is basic and sometimes unclear	Poor, unclear, or disorganized explanation	
5	**Question & Answer (Q&A)**	Answers all questions confidently and accurately	Answers most questions correctly	Answers some questions with hesitation some questions with hesitation	Unable to answer questions correctly	
				TOTAL	20	

Labwork #1 - Shodan Searches

Objectives

Shodan is a search engine for IoT devices that was developed by John Matherly in 2009. Shodan can discover all types of internet-connected “things", from mobile phones, to smart appliances, to power plants. It is a powerful tool to determine what devices are currently connected to the network and how they are connected.

Create a Shodan user account and register for an API key

Use the Shodan website to search for vulnerable IoT devices

Use Shodan from the CLI to perform a search

Background / Scenario

IoT devices are in wide usage. They are created, installed, and maintained by governments, businesses, and homeowners. These devices are not usually hardened by the manufacturer. It is the responsibility of the end-user to ensure that these devices do not introduce additional risks to network security.

You can perform some Shodan searches without obtaining a subscription. More extensive searches require a paid subscription.

In this lab, you will conduct a Shodan search for vulnerable devices within your private network, as well as within a defined IP address range. As with most tools that you are using in this course, only scan or access networks that you own or have permission to access.

Required Resources

Kali VM customized for Ethical Hacker course

Internet access

Instructions

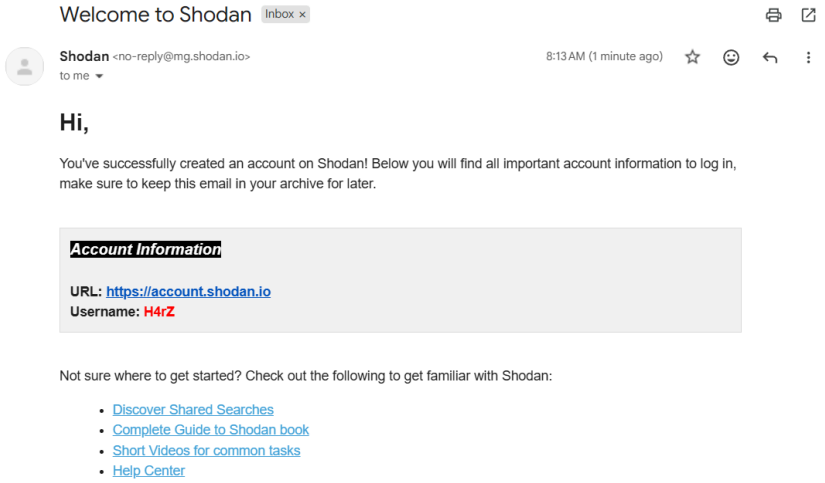
Part 1: Create a Shodan Account and Register for an API Key

Step 1: Register for a Shodan account.

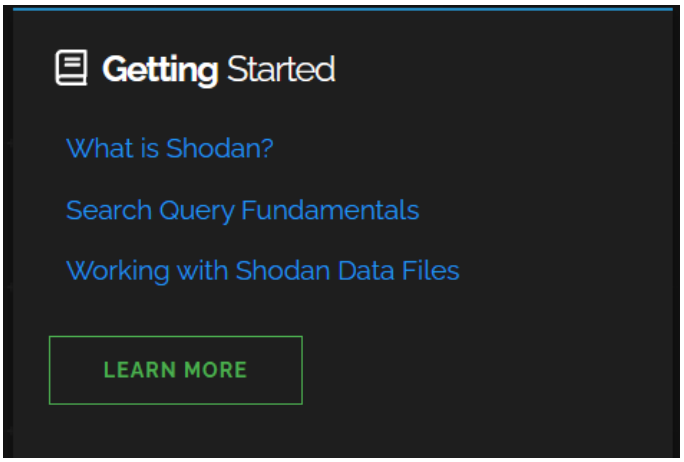
a. Login to your Kali Linux VM.

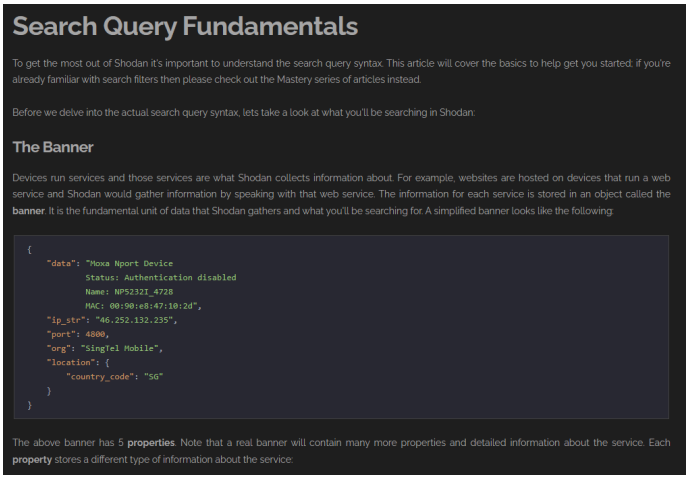
b. Open the Firefox browser and navigate to https://www.shodan.io/.

c. Click the **Login** button at the top right. On the next screen, click the **Register** button on the menu bar. Fill in your information to create a Shodan account. You will receive an email to activate your account.



d. When your registration is complete, log in to your Shodan account. This is a free account that has several restrictions, including the number of results that will be displayed from each search. Log in and go to the Shodan home page. Review the **Getting Started** section, especially the **Search Query Fundamentals** link.





Question

According to Shodan, what is the fundamental unit of data it gathers?

According to Shodan.io, the fundamental unit of data it gathers is the banner. A banner is the textual metadata that a device or service sends back to a client.

Example :

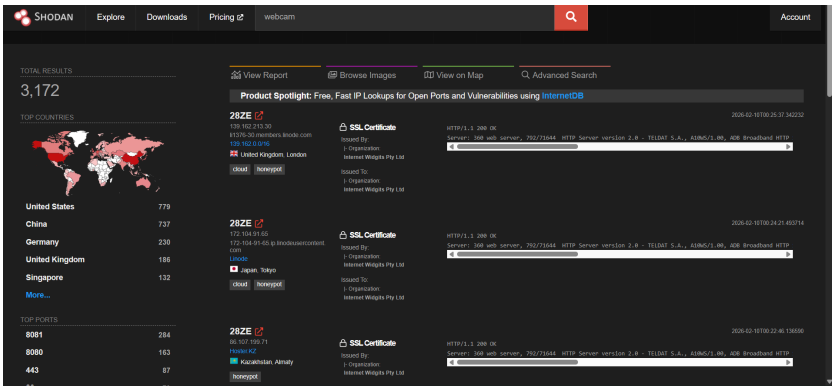

```
{
    "data": "Moxa Nport Device
             Status: Authentication disabled
             Name: NP5232I_4728
             MAC: 00:90:e8:47:10:2d",
    "ip_str": "46.252.132.235",
    "port": 4800,
    "org": "SingTel Mobile",
    "location": {
        "country_code": "SG"
    }
}
```


Part 2: Use the Shodan Website to Search for Vulnerable IoT Devices

Step 1: Use the Shodan search bar to discover IoT devices.

a. On the Shodan home page, enter **webcam** in the search bar near the top of the screen and press enter.

b. A page displaying search results will appear. On the left side of the screen are summary statistics. The statistics show the total number of device banners that include the term “webcam”, the top countries where the results were found, the top organizations, top products, and top operating systems. You can view up to 10 results without a Shodan login. Registered users can access 50 results for free. Additional services are available with a paid subscription.

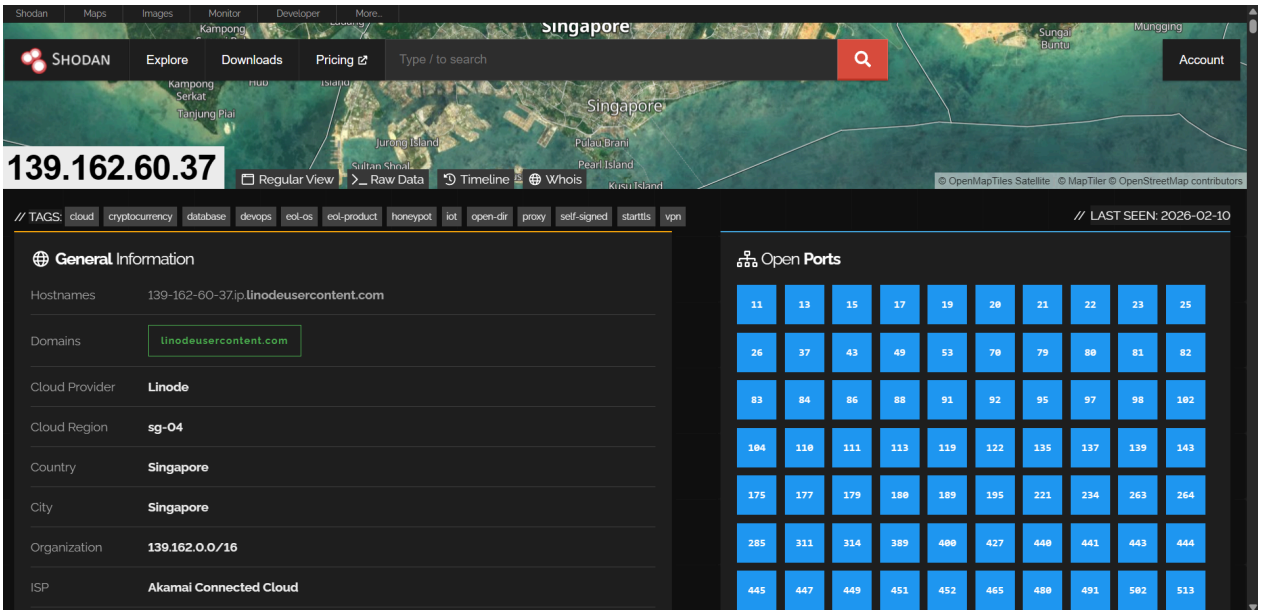


Question

What is the top country listed with web cams found by Shodan?

United States.

c. Click one of the IP addresses listed in the search results. A page with more detailed information opens. At the top of the page, there is a map that shows the approximate location of the search result that you selected. Explore the information for several of the devices that were discovered.



Question:

What information is contained in the General Information section?

There are hostname, domains, cloud provider, ISP as well as the country information. Below is also a list of vulnerabilities found on the server. While on the right side there is a list of open ports.

Questions:

What ports are open on the IP address that you selected?

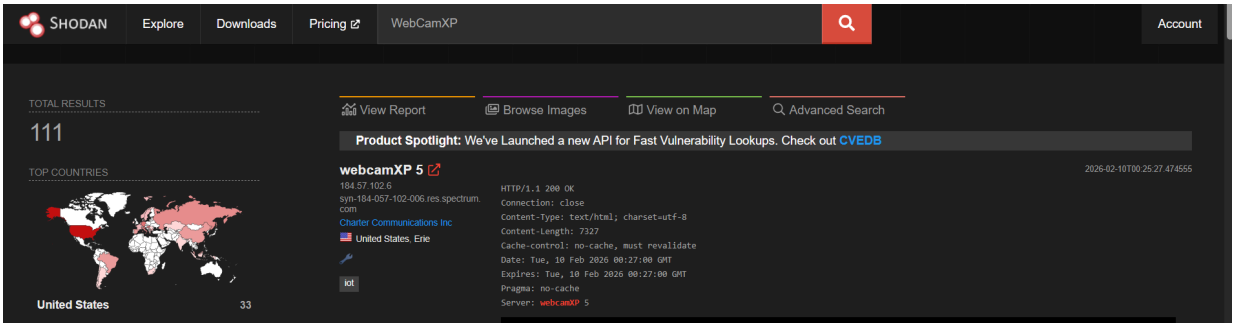
11 - 9209.

What information is available for the open ports?

The available information indicates that this IP address is a **honeypot** that intentionally exposes a massive variety of open ports.

Note: Not all the devices discovered are actually webcams. They are devices that have the word “webcam” somewhere in their service banners.

d. Search the web for “vulnerable webcam manufacturers.” Frequently, device banners will name the manufacturer of the device. Try searching for some manufacturer names in Shodan. From the results, you can refine your search results, sometimes with specific manufacturer’s model numbers. In addition, search for default logins used by camera models. It is possible that the owner of the camera did not change the default password. **DO NOT** attempt to login to devices that you do not own or have permission to access.



Step 2: Use Shodan filters to refine the results.

Shodan provides a method to filter your search results using the syntax ***filter:value*** with no spaces. If the value contains spaces, such as **city:"los angeles"**, you must enclose the value in double quotes. Some of the most popular search filters are:

`country:XX` Searches for a 2 digit country code

`city:city-name` Searches for a city by name

`region:region-or-state-name` Searches for a specific state or region

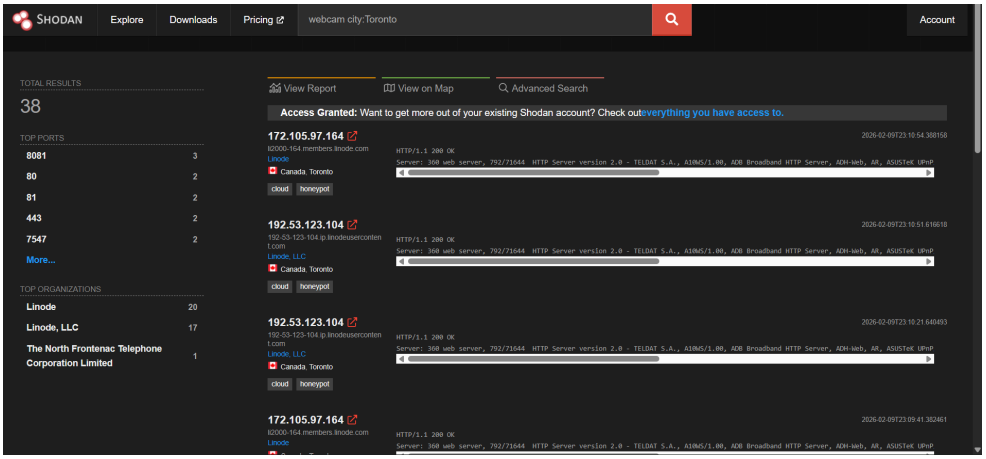
`product:product-name` Searches for a specific product by name

`version:``XX` Searches for a specific product version

`vuln:XX` Searches for vulnerabilities that match a specific CVE number

a. Enter a filter on the Shodan search bar. This example returns all the devices with “webcam” in a banner that Shodan finds in the city of Toronto.

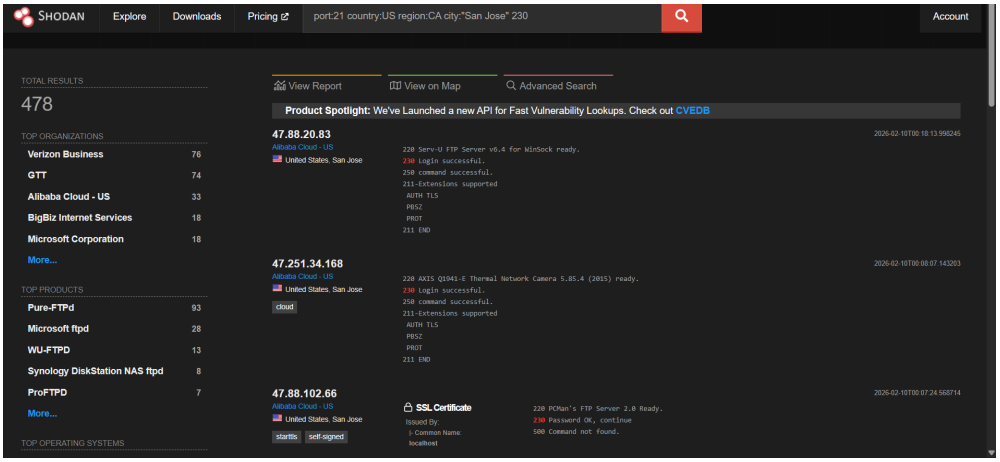
`webcam city:Toronto`



b. A common configuration issue found on the internet is FTP servers that permit anonymous logins. Use the search string to find the FTP servers in San Jose, California.

`port:21 country:US region:CA city:"San Jose" 230`

This search uses the standard FTP TCP port 21, with location filters, and a text search for 230. 230 is the FTP successful login response code.

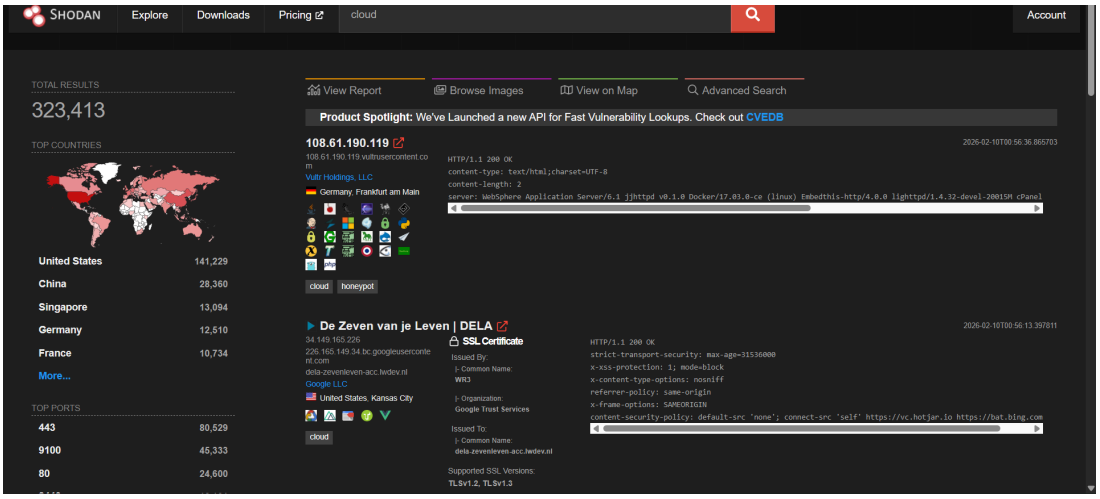


Question

How many FTP servers did Shodan find in San Jose that permitted anonymous logins?

478.

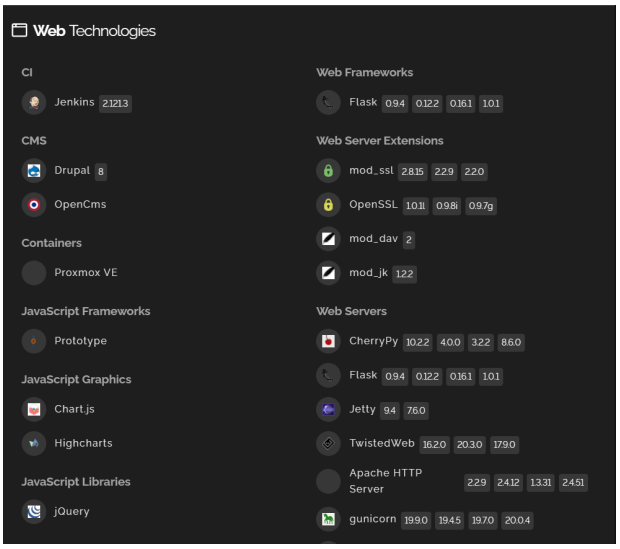
c. Shodan searches can identify cloud applications and possible honeypots. Browse the search results from your queries to find results labeled **cloud** or **honeypot**.



d. Click one of the results labeled **cloud** to open the details page.

Question

What additional information is contained in the General Information section when compared to the result you recorded in Step 1c?

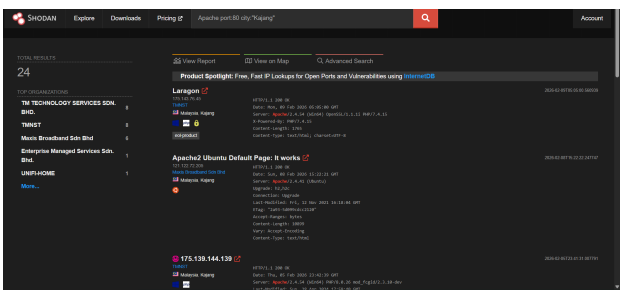


There is an additional **Web Technologies** section in the General information Section for this IP Address. Compared to Step 1C where it only has the **Vulnerabilities** section.

Step 3: Use Shodan to search for a specific product or service.

You can use Shodan to search for a specific product, such as Apache servers open on port 80. Formulate a query to find the Apache servers in your city.

`Apache port:80 city:"`*`your-city`*`"`



Part 3: Use Shodan from the CLI to Perform a Search

Step 1: Initialize Shodan and perform a search.

a. Find and copy your API key by selecting **Account > Overview** from the top right of the Shodan web site screen. Highlight the API key shown above the QR code, right-click the selection and select **Copy**. Make note of your key for future use.

b. Shodan is a Python library that is installed in Kali by default. Open a Kali terminal window.

c. At the prompt, enter the command **shodan init** and right click and select **Paste Selection** to paste the API key into the terminal. Your API key should appear at the end of the command.

```
┌──(root㉿kali)-[/home/kali]
└─$ shodan init <paste your API key here>
```

This command should return the string "**Successfully initialized**".

d. Enter the **shodan -h** command to display the list of Shodan commands available from the command line.

```
┌──(kali㉿kali)-[~]
└─$ shodan init kgJmwuAG5abgeAnbRipjWyYIlK2ElBcR
Successfully initialized

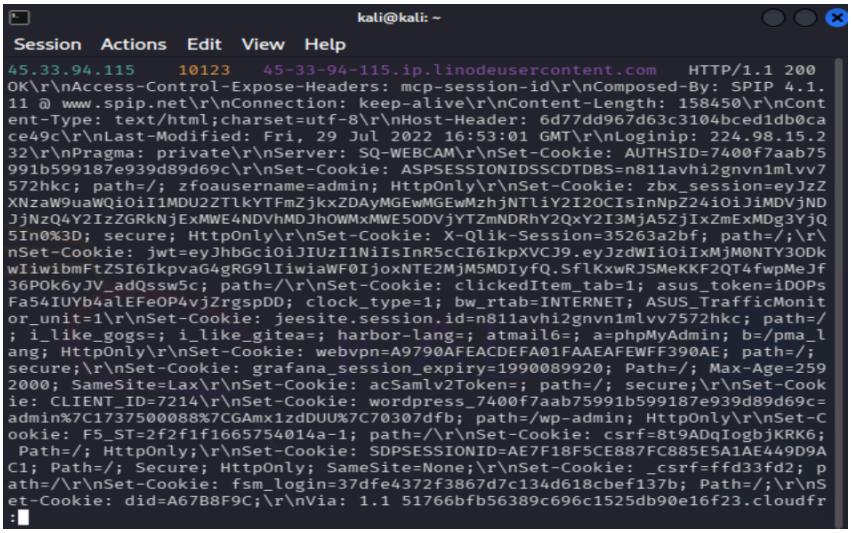
┌──(kali㉿kali)-[~]
└─$ shodan -h
Usage: shodan [OPTIONS] COMMAND [ARGS]...

Options:
  -h, --help  Show this message and exit.

Commands:
  alert       Manage the network alerts for your account
  convert     Convert the given input data file into a different format.
  count       Returns the number of results for a search
  data        Bulk data access to Shodan
  domain      View all available information for a domain
  download    Download search results and save them in a compressed ...
  honeyscore  Check whether the IP is a honeypot or not.
  host        View all available information for an IP address
  info        Shows general information about your account
  init        Initialize the Shodan command-line
  myip        Print your external IP address
  org         Manage your organization's access to Shodan
  parse       Extract information out of compressed JSON files.
  radar       Real-Time Map of some results as Shodan finds them.
  scan        Scan an IP/ netblock using Shodan.
```

e. Execute the same search at the CLI that you did in the web search bar to view webcams that Shodan finds. At the CLI, you must enter the **shodan search** command before specifying the search criteria.

```
┌──(kali㉿kali)-[~]
└─$ shodan search webcam
```



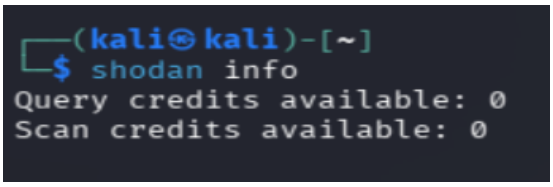
The output of the command is unformatted text. The IP addresses of the devices that Shodan finds are highlighted along with the port and the device name. Press **q** to quit and return to the CLI prompt. Shodan CLI commands can be written into Python scripts to automate search and scanning functions.

Note: Searching with filters is not available with a free API key.

Step 2: Execute other Shodan CLI commands.

a. Not all commands listed are available in the free version of Shodan. The **shodan info** command will show how many credits that you currently have to perform searches or scans.

```
┌──(kali㉿kali)-[~]
└─$ shodan info
Query credits available: 0
Scan credits available: 0
```



For paid subscriptions, the available credits will reset each month. There are subscriptions available for a cost that permit unlimited queries and scans.

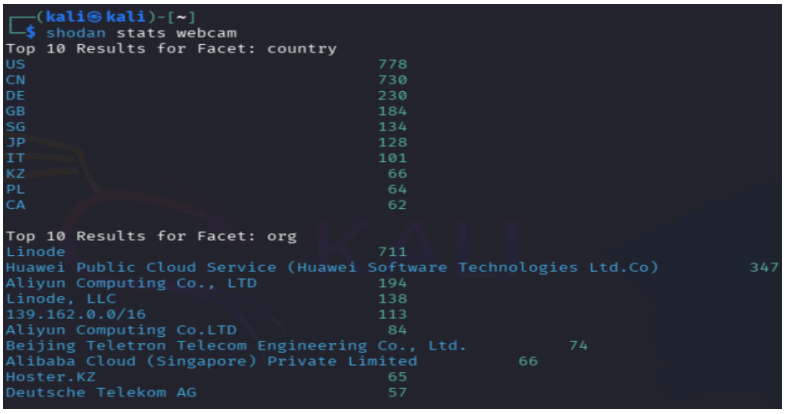
b. Use the **shodan myip** command to find the registered IP address that corresponds to your device.

The IP address returned is the source IP address that will be added to any packets sent from your device to a destination on the internet.



c. Another useful command is the **stats** command. It will return the summary information about a query, similar to what is displayed on the results page in the web version. To get the summary information, enter the stats command and the search query that you want to view the results.

```
┌──(kali㉿kali)-[~]
└─$ shodan stats webcam
```



This query returns the summary statistics for the webcam search.

Reflection Question

Shodan can provide a wealth of information about systems and devices that are connected and communicating on the internet. What features of Shodan are especially valuable for IT administrators?

Shodan is especially valuable to IT administrators for its network monitoring and alerting features, which allow them to gain an "outside-in" view of their infrastructure to identify exposed assets, verify firewall effectiveness, and detect unpatched vulnerabilities.